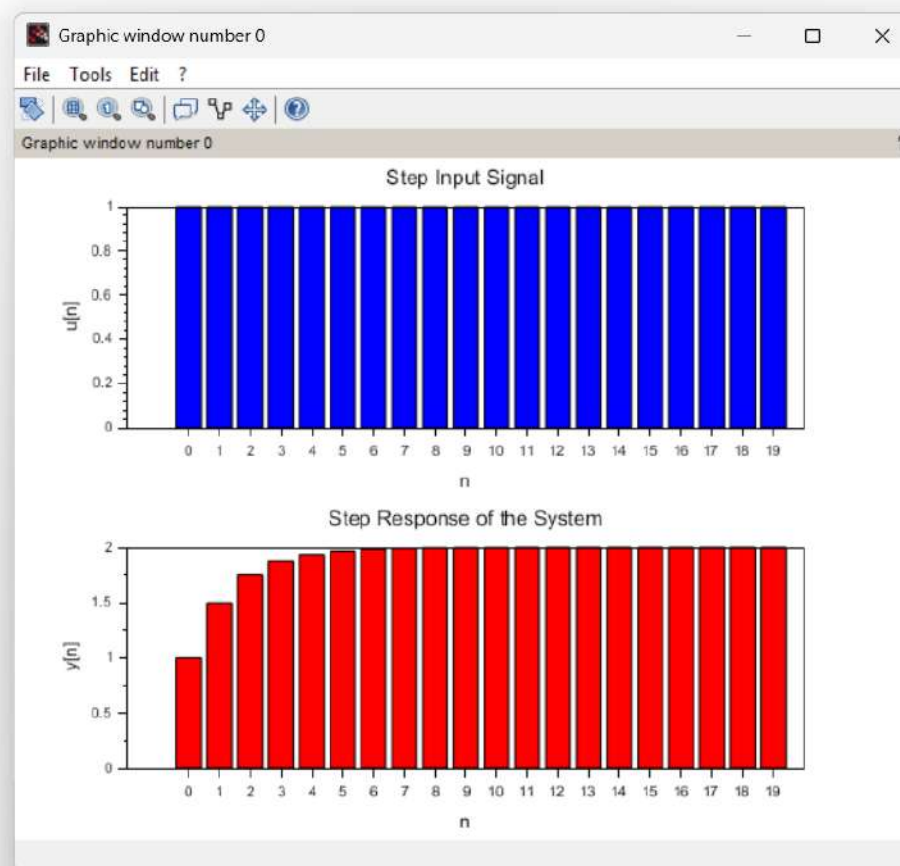


```

1 // Test Case: Step Response of a Discrete-Time System.
2 // Define the length of the signal.
3 N = 20;
4 // Length of the signal.
5 // Define the input signal (step function).
6 u_n = ones(1, N);
7 // Step function.
8 // Define the system parameters.
9 b = [1];
10 // Numerator coefficients of the system (input).
11 a = [1, -0.51];
12 // Denominator coefficients of the system (output).
13 // Compute the system's response to the step input.
14 y_n = filter(b, a, u_n); // System output for step input.
15 // Define the time index for the signals.
16 n = 0:N-1;
17 // Time index.
18 // Plot the step input and system response.
19 clf; // Clear the current figure.
20 // Plot step input.
21 subplot(2,1,1);
22 bar(n, u_n, 'b');
23 xlabel('n');
24 ylabel('u[n]');
25 title('Step Input Signal');
26 // Plot system response to step input.
27 subplot(2,1,2);
28 bar(n, y_n, 'r');
29 xlabel('n');
30 ylabel('y[n]');
31 title('Step Response of the System');
32

```





EXP_5.4 SS.sci (E:\EXP_5.4 SS.sci) - SciNotes

EXP_5.4 SS.sci

```
1 // Test Case: Analysis of a Discrete-Time System.
2 // Define the input signal (unit impulse).
3 N = 20;
4 // Length of the signal.
5 x_n = zeros(1, N);
6 x_n(1) = 1;
7 // Initialize input signal.
8 // Unit impulse at n=0.
9 // Define the system parameters.
10 b = [1];
11 // Numerator coefficients of the system (input).
12 a = [1, -0.5];
13 // Denominator coefficients of the system (output).
14 // Compute the impulse response using the filter function.
15 h_n = filter(b, a, x_n); // Impulse response of the system.
16 // Define a different input signal (e.g., a step function).
17 u_n = ones(1, N);
18 // Step function.
19 // Compute the system's response to the step input.
20 y_n = filter(b, a, u_n); // System output for step input.
21 // Define the time index for the signals.
22 n = 0:N-1;
23 // Time index.
24 // Plot the impulse response and system response.
25 clf; // Clear the current figure.
26 // Plot impulse response.
27 subplot(2,1,1);
28 bar(n, h_n, 'b');
29 xlabel('n');
30 ylabel('h[n]');
31 title('Impulse Response of the System');
32 // Plot system response to step input.
33 subplot(2,1,2);
34 bar(n, y_n, 'r');
35 xlabel('n');
36 ylabel('y[n]');
37 title('Response of the System to Step Input');
38
```

